

TrES-1b

R-2103

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_110526 · created 2026-08-10T23:35:40+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455707.82432 +/- 0.00061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1226 +/- 0.0058
Transit depth (Rp/Rs)^2	1.5 +/- 0.14 [%]
Orbital Inclination (inc)	89.94 +/- 0.53
Airmass coefficient 1 (a1)	1.1573 +/- 0.0094
Airmass coefficient 2 (a2)	-0.1349 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	None
Optimal Aperture	4.57
Optimal Annulus	14.96
Transit Duration (day)	0.10274 +/- 0.00075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1226 $\pm$ 0.0058 vs 0.1358 (deviation 2.03 σ)
Duration fit vs published	0.10274 d vs 0.1042 d (deviation 1.7 σ)
Mid-time O-C	-1.9 min
Depth SNR	18.8
Residual scatter	0.81 %

Light curve

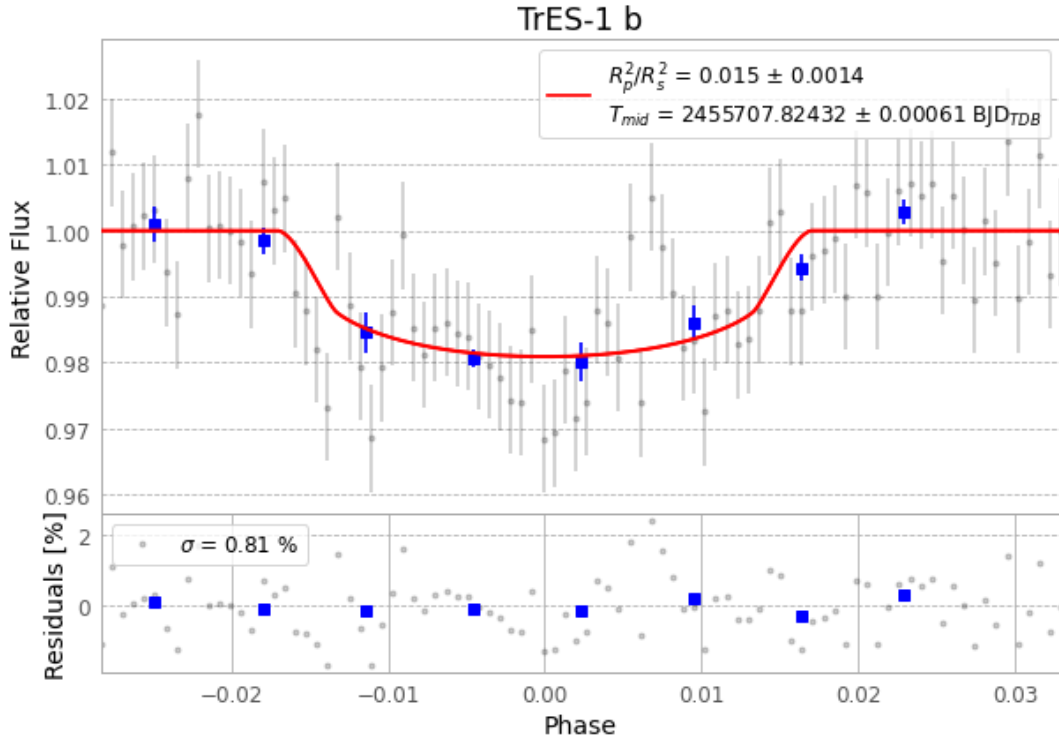


Figure 1 — FinalLightCurve_TrES-1 b_2011-05-25.png (SHA-256 b46796849844bf9c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [244, 212], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 85a0fdb97801e678...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

90 FITS frames (59 MB), TRES-1110526053913.FITS → TRES-1110526100613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-110526.log (dca92cd4d058...), FinalLightCurve_TrES-1 b_2011-05-25.png (b46796849844...), FinalLightCurve_TrES-1 b_2011-05-25.csv (209f04f303b9...)

Compute resources

Wall time 197.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:35Z.